

Capitalizing on Empirical Evidence During Agile Adoption

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Abstract—Despite high expectations, agile methods have not always produced the desired results. The suitability of various agile practices depends on many situational factors, such as team size, project complexity, individuals competence, etc. As agile practices have become increasingly widespread, there is a growing body of literature that reports on empirical experiences, including factors that contribute to successful or problematic adoptions. However, this considerable body of knowledge is hard for the busy practitioner to digest, synthesize, and put to practice for the specific context at hand. Our previous work [1] introduced a repository of agile methods that provides evidential knowledge about the needed conditions (requisites) and promised benefits (objectives) of various Agile Method Fragments (AMFs). The knowledge was gathered through systematic review of empirical studies which investigated the enactment of agile methods in various project situations. This paper reports the use of repository in a software process improvement initiative in one of the R&D units of Ericsson Company in Italy.

Keywords – Agile Methods, Empirical Software Engineering, Method Engineering, Software Process Improvement

INTRODUCTION

ONE OF the primary concerns of project managers, who want to move to agile style of software development, is to figure out the extent that the promises of the new method can be attained in their particular organization setting. Transition to agile software development, or in general to any new development method, is usually aligned with doubts and uncertainties. This cautious attitude is due to a number of reasons, including the inertia of organizations in keeping the current process; the failure experiences of other organizations in moving to agile; the lack of competence for implementing agile; or the suspicion of managers to the claims of agile methods.

Upon the introduction of any new method, a number of advantages will be claimed for it, and certain conditions will be described to justify the feasibility of method for organizations or enhance its effectiveness. However, after the actual trial of that method in different software organizations, solid evidences will be proposed that had tested the attainability of the claimed benefits and necessity (or completeness) of referenced conditions. Some of the claimed advantages might turn to be highly dependent on the presence of certain preconditions; or even some new advantages, disadvantages, and preconditions might be introduced for the method.

The widespread adoption of agile methods by software companies, has provided a suitable test bed for research community of software engineering, who are interested in

empirical investigation of agile methods. Every year, a high volume of empirical studies on the deployment of agile methods in different organization settings will be published, mostly investigating the effects of those methods on the organizational goals, and the significant criteria for their success or failure. For instance, a simple title-based search in the IEEE Explorer for "Pair Programming" returns over 50 results, mostly reporting on empirical studies.

Despite the considerable volume of published empirical studies on agile methods, there is no systematic approach for accessing this body of knowledge. In our previous work, we introduced a repository of Agile Method fragments (AMFs), which provides information about their objectives and requisites [1]. The definition of a *method fragment* follows [2, 3] explained as a piece of a development method, which can be classified as product, process, or producer fragments. For example, "Pair Programming" is a method fragment, which is a piece of XP development method [4]. The information stored in that repository is evidence-based, as they have been extracted through systematic literature review of published empirical studies on agile methods. For each AMF, the repository contains two datasets, the *objectives dataset* and the *requisites dataset*. The objective dataset introduces the objectives that have been reported to be affected (in either positive or negative ways) by the enactment of an AMF; and the requisites dataset introduces the preconditions that have been reported to be needed for the successful enactment of an AMF in a software project.

This paper reports one of our experiments of using this repository in an industrial project. The repository's knowledge of AMFs was used during a Software Process Improvement (SPI) initiative in one of the R&D units of Ericsson Company in Italy. One of the objectives of this initiative was to change the software development style with respect to agile principles, and integrate a number of AMFs into the current process of software development. We took advantage of this repository in order to gain a realistic view of the benefits and drawbacks of agile methods, and make informed decisions while selecting a set of AMFs for this organization.

I. BACKGROUND: THE REPOSITORY OF AGILE METHOD FRAGMENTS

A. Data Collection

The research method for populating the repository is the systematic review [5] of published empirical studies on agile methods. Our primary research goal is to identify the actual objectives and requisites of different AMFs.

We started the research by selecting 20 agile method fragments for consideration. For each method fragment we looked up published empirical studies that explained the enactment of that method fragment in a particular project/organization. We searched the major collections of computer science publications, including IEEE Explorer, ScienceDirect, SpringerLink, and ACM Digital Library. We used search queries that contained the name of the method fragment, the name of the originating agile method, and some phrases for retrieving empirical studies, such as: Case study, Experiment Report, Ethnography, or Survey.

In addition to published empirical studies in academic journals and conferences, we also considered articles from agile web-forums as complementary sources for our reviewed empirical studies. Every selected paper or article was then evaluated with respect to the following items: 1- reliability of reported data; 2- reproducibility of study; 3- relevance to the research questions; 4- quality of situation description (context for enacting AMF). Only publications that meet all of the above criteria are used as sources, and are listed as references.

The extracted data from reviewed empirical studies were then analyzed and aggregated as units of knowledge for every method fragment. Data analysis was performed in order to:

- Clarify the contributions of method fragments to objectives. The contribution relations were represented qualitatively, using a five-value scale (--, -, ?, +, ++). Negative contributions represent the cases where a method fragment was reported to strongly deny (--) or somewhat deny (-) an objective; unknown contributions (?) represent cases where a study reported that an AMF does not have any significant contribution to a particular objective; and positive ones represent the cases where a method fragment was reported to strongly satisfy (++) or somewhat satisfy (+) a quality objective.
- Resolve the naming problems that were due to synonyms or homonyms. Empirical studies use different terms and taxonomies to describe process objectives. We analyzed the extracted data to harmonize the various terminologies of reviewed studies. shows a subset of harmonized taxonomy of AMF objectives.
- Classify the method objectives and requisites into two-level categories. The extracted objectives (or requisite) were classified either as major or minor categories. Minor categories are considered as subcategories of major ones, and represent more delicate aspects of a major quality objective (or requisite). Although the objectives and requisites of a method fragment can be detailed with further levels of refinement, for the sake of simplicity, we consider only two levels for the current design of the repository. and show two examples of this categorization.

The analyzed data were then aggregated with what had been collected previously. The knowledge of each method fragment was aggregated in two tables, one for its objectives and the other for its requisites. Data aggregation for AMF was either in the form of adding a new objective/requisites; or providing a new evidence for existing contribution relation. The final stage of data

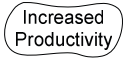
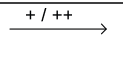
aggregation was to make generic snapshots, which are two dimensional tables that show the identified (and unidentified) relations of all method fragments to all objectives or requisites. Generic snapshots facilitate the comparison of method fragments, and provide insights for further empirical studies to uncover the potential relations between agile method fragments and process objectives or requisites. shows a sample generic snapshot.

B. Data Visualization

An appropriate modeling paradigm for visualizing the collected data of this study should be capable of representing method objectives and requisites, also the contribution relations of AMFs to their objectives and their dependency relations to their requisites. These features have been well-addressed in Goal-Oriented Requirements Engineering (GORE), in which goal models are used to express the intentional elements of a modeling domain, and the contribution of domain objects, including activities and resources, to the domain goals [6]. In this research we used the i^* goal-oriented modeling framework for visualizing the collected knowledge of AMFs.

Table 1 describes a subset of i^* modeling elements, which were used for visualizing the objectives and requisites of method fragments. Fig. 1 shows an example of AMF visualization. Further description of our approach in modeling method fragments, and their application in process engineering are presented in [7].

TABLE 1.
THE USE OF i^* MODELING NOTATION FOR VISUALIZING METHOD FRAGMENTS

i^* Modeling Element	Description	Graphical Notation
Task	Represents the enactment of an AMF, and places at the center of each model	
Softgoal	Represents both AMF objectives and requisites	
Contribution Relation	Visualizes the contribution relation of an AMF to its objectives	
Decomposition Relation	Visualizes the dependency relation of an AMF to its requisites	

II. THE REPOSITORY IN PRACTICE

We used the structured information of agile method fragments in a software process improvement project in one of the R&D units of Ericsson company. Managers of the R&D unit had decided to improve the efficiency of their development process by moving to agile. However, they were hesitant on the suitability of the AMFs that were suggested to them, and were interested to know more details about AMFs' added benefits and needed conditions. This information was easily accessible through our repository of agile method fragments. Considering the candidate process that the R&D unit was about to adopt, which was mainly based on Scrum [8], we used the repository to extract the knowledge of the following method fragments: (1) Scrum Team Structure,

(2) Sprint Planning, (3) Short Release, (4) Daily Scrum Meeting, (5) On-Site Customer.

The information presented about the objectives and requisites of the candidate method fragments, along with the situational evidences that were provided as well, helped the managers to decide on the suitability of their candidate AMFs. In the following subsections as a sample AMF, we present the information that we collected for the AMF "On-Site Customer". The complete set of our tables are now published as a repository of method

fragments, and publicly accessible at <http://www.cs.toronto.edu/km/MFR/MF.htm>.

A. On-Site Customer: Objectives

shows a subset of objectives of the agile method fragment "On-Site Customer". It presents different types of contributions that this AMF might make to its objectives, along with references to the empirical studies which have investigated these issues. Besides, it briefly

TABLE 2. A SUBSET OF THE OBJECTIVES OF THE AMF, "ON-SITE CUSTOMER", ALONG WITH SITUATIONAL EVIDENCES FOR ITS CONTRIBUTIONS

Major Objective	Minor Objective	Contribution	Study	Situation
Enhanced Communication	Facilitated Communication with business people	++	S39, S40	In General
		-	S36, S56	large project - overloaded customer, who could not play the role of all stakeholders
	Better understanding of customer needs, also technical concerns of customer	++	S37, S39	In General
Enhanced Collaboration	Higher participation of user in planning	+	S39	customers were trained to have a minimum required planning knowledge
	Higher participation of user in defining Test Cases	++	S39	customers were trained to have a minimum required technical knowledge
Improved Effectiveness	Reduced need for requirement models and documentation	++	S56	
		--	S56	Large complex systems - enterprise systems that need to ensure alignment with business goals
	Balancing potential conflicts between customer needs and developer's	++	S39	In General
	Fast resolvent of requirement issues	++	S34, S36	In General

describes the situations in which stated contributions had occurred. For instance, it shows that Pair Programming, in general cases (represented as Default), contributes strongly and positively to the objective "Reduced Defects in Code". It also provides evidence from a controlled experiment, which qualifies this contribution with more specific situational information (complex system with junior developers).

The contributions of a method fragment to an objective can be classified in two categories. The first category is for objectives that a method fragment makes situation-independent contributions to them, i.e. the change of context does not sharply change the contribution type of a method fragment to them. For example, in all of the reviewed empirical studies reported that Pair Programming makes positive contribution to the objective "Faster Problem Solving". But, there exist another category of objectives for which the contribution on AMF is situation-dependent, i.e. the change of context can alter the type of contribution. For instance, the default contribution of "Pair Programming" to the objective "Be on-time to market" is positive, but [9] reports that Pair Programming in the context of a large project with limited number of developers and high market pressure can adversely contributes to that objective.

A. Agile Method Fragments: Requisites

As an example of evidential knowledge about requisites, shows a subset of major and minor requisites of the agile method fragment "Daily Scrum Meeting". For each requisite, it presents a number of empirical studies which had reported the impact of that requisite on the success or failure of daily scrum meetings. This table

briefly describes the significant situational factors of the referenced studies, and mentions the achievement (or denial) status of requisites in those situations. The achievement status of requisites is specified either as: Denied (D), Partly Denied (PD), Partly Satisfied (PS), or Satisfied (S).

B. Agile Method Fragments: Visualization

Fig. 1 shows a sample visualization of an agile method fragments. It depicts some of the objectives that pair programming contributes to them. For instance, it shows that pair programming contributes positively to the objective "Better Time to Market". We used four types of contribution relations from the *i** modeling framework, in order to represent the contribution of an AMF to its objectives: "++" representing strong positive contributions; "+" representing normal positive contributions; "-" representing normal negative contributions; "--" representing strong negative contributions. Fig. 1 also shows the dependency of the AMF pair programming to some of its requisites, e.g. "Effective Collaboration of Pairs".

III. DISCUSSION AND FUTURE WORK

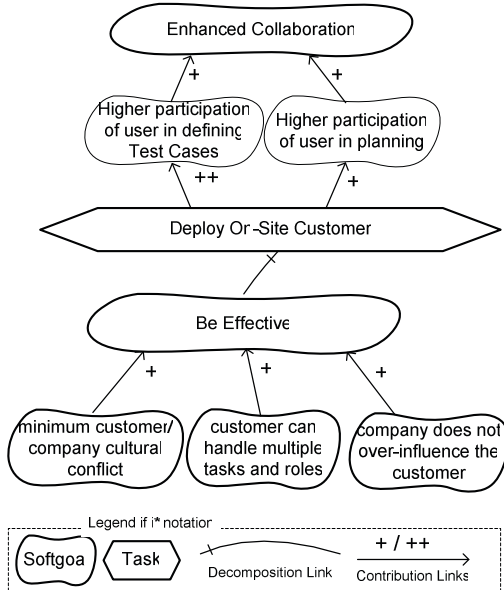
So far the repository has been populated with the knowledge of 20 agile method fragments. The website for this repository is being regularly updated to reflect the latest findings of empirical software engineering about agile method fragments. The plan is to expand the repository to non-agile method fragments. Work has started to collect and analyze method fragments for requirements engineering, agile as well as non-agile.

TABLE 3.
A SUBSET OF THE REQUISITES OF THE AMF, "ON-SITE CUSTOMER", AND THEIR ACHIEVEMENT STATUS IN A NUMBER OF REFERENCED EMPIRICAL STUDIES

Major Requisite	Minor Requisite	Achievement	Study	Situation
Good Collaboration	avoid under utilization of customer	PD	S34	from 100% availability of customer, only 20% was necessary
	minimized technological gap between customer and developers	PS	S39	training the customer (as needed),
commitment	customer be available when needed	D	S33	lack of on-site customer
be effective	minimum customer/company cultural conflict	PD	S33	
	customer can handle multiple tasks and roles	D	S36, S37	Large project (S6) - customer could not play the role of all stakeholders
		S	S39	Team of Customers, rather than one customer were on-site
		S	S39	Classical stakeholder analysis had been performed, before working with on-site customer
	Company does not over-influence the customer	D	S38	Development team could easily convince the customer, without actually implementing some of the user stories
	Customer does not over-influence the team	D	S38	when customer noticed that he can influence the team, started to complain about anything and everything
		S	S39	customer (team) was trained on how, and when to represent their idea, besides he was asked to think of "...so that..." in order to have a good reason for his demands

The results of using this repository in an SPI initiative show that we are able to reuse the knowledge stored in this repository, in order to analyze the potential impacts of deploying a set of AMFs on the strategic goals of organization. Our aim is to develop a comprehensive SPI framework, drawing on the evidential knowledge of the repository. This framework includes a process evaluation technique, which makes use of the pre-constructed goal models to investigate the suitability of a set of AMFs for a particular project/organization [5].

FIG. 1.
VISUALIZATION OF THE AGILE METHOD FRAGMENT "ON-SITE CUSTOMER", FOR A SUBSET OF ITS OBJECTIVES AND REQUISITES



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